Customer NPS Prediction for a Telecom Operator

Take-Home Challenge — Technical Write-Up

Prepared for Artefact

1. Approach and Problem Framing

The brief asks for a model that predicts a customer's NPS category (Detractor, Passive, Promoter) from account and behavioural data, so a retention team can prioritise outreach toward likely Detractors without waiting for a survey response — only around 15% of customers ever answer one. Two properties of this problem shaped every decision documented below: NPS is ordinal (a Detractor predicted as Promoter is a materially worse error than Detractor predicted as Passive), and the label itself is derived, not observed — this dataset provides a five-point Satisfaction Score, not a direct NPS answer, so target construction had to be treated as a first-class decision rather than a preprocessing footnote.

Three model families were trained and compared — a class-weighted logistic regression baseline, a gradient boosting classifier, and an ordinal logistic model — specifically to let evidence, not assumption, answer the brief's central question of how to frame an ordinal target. No single family dominated on every metric, and the final model choice changed once during the project as a direct result of measuring calibration and lift rather than stopping at classification accuracy. That reversal is documented explicitly in Section 4 rather than smoothed over, because it is itself a finding: which model is "best" here depends on which part of the business problem is being optimised.

2. Target Construction

NPS category was derived from Satisfaction Score using two candidate mappings, compared rather than assumed. The literal baseline mapping suggested in the brief (score 5 → Promoter, 4 → Passive, 1–3 → Detractor) produces a 58.3% Detractor rate — implausibly high for a real NPS distribution, and a direct artifact of collapsing three of five satisfaction levels into one class.

The refined mapping treats a score of 3 as genuinely ambiguous and resolves it using each customer's realised churn outcome: a score-3 customer who subsequently churned is labelled Detractor, one who stayed is labelled Passive. This produces a 26.5% Detractor rate that matches the dataset's true churn rate almost exactly — a strong validation signal, since the two quantities were derived independently. The two mappings disagree on 31.7% of customers, so this was not a cosmetic choice; both are retained in the repository for sensitivity comparison.

Leakage handling: Satisfaction Score, Churn Score, Churn Value, Churn Label, and Customer Status are excluded from every model's feature set. This project also caught and fixed a subtler leak: an intermediate column created for the mapping-sensitivity comparison ("NPS_baseline") was initially left in the feature table by omission, where it dominated every SHAP analysis at roughly double the importance of the next feature, since it agrees with the true target on 68.3% of rows. The fix — and the discipline of asserting its absence at the top of every downstream notebook — is now part of the pipeline, not just a one-time correction.

3. Data Preparation and Feature Engineering

The two source files (a 21-column demographics/services table and an 11-column status table) were merged on customer ID and validated one-to-one — both cover the same 7,043 customers with no orphans on either side. TotalCharges arrives as a string with 11 blank values, all corresponding to customers with zero tenure who have not yet been billed; these were set to 0.0 rather than imputed, since a generic imputation would misrepresent a real, explainable case. Two constant columns (Count, Quarter) were dropped as uninformative.

Twenty-four features feed the production model: demographics, contract and account attributes, service subscriptions, billing amounts, and four engineered features — a service count, a charges-per-service ratio (to separate a customer paying a lot for few services from one paying a similar amount for many), a household-size proxy, and an automatic-payment flag. Geographic features (ZIP code, latitude/longitude) were scoped out: the relevant IBM Telco Location file was not available for this project, and this is stated here as a limitation rather than papered over with a placeholder.

The train/test split (80/20, stratified on NPS category) preserves class proportions between folds, but it cannot simulate the real production gap between survey respondents and the silent majority, since Satisfaction Score happens to be populated for 100% of customers in this dataset rather than the 15% the brief describes. This is a genuine limitation of the available data, not a solved problem, and is flagged as a monitoring priority in Section 7.

4. Modelling and Evaluation

All three model families comfortably outperform a naive majority-class baseline (Macro-F1 0.242), confirming the features carry real signal.

Model	Macro-F1	Balanced Acc.	QW Kappa	Far-error rate	Top-20% lift
Logistic Regression	0.513	0.592	0.346	10.0%	47.6%
Gradient Boosting	0.514	0.533	0.290	8.5%	46.0%
Ordinal (LogisticAT)	0.475	0.481	0.365	2.6%	42.0%

Far-error rate is the share of predictions two classes away from the truth (Detractor predicted as Promoter, or the reverse) — the costliest kind of mistake for this use case. Top-20% lift is the share of true Detractors captured within the top 20% of customers ranked by predicted risk, which mirrors exactly how a retention team would act on this model's output: call down a ranked list.

No model wins on every column. The ordinal model has the best quadratic weighted kappa and by far the fewest severe misclassifications, which made it the initial choice for production. Measuring calibration and lift in a follow-up evaluation changed that decision: none of the three models is well-calibrated in an absolute sense (class weighting, necessary to get any signal on the minority classes, distorts predicted probabilities as a side effect), and on lift — the metric that most directly answers "how many Detractors does the retention team reach by calling the top of the ranked list" — logistic regression leads clearly. Since the brief's stated business objective is outreach prioritisation, logistic regression is the model selected for production. This reversal is recorded here explicitly: it is evidence that metric choice, not just model choice, determines the answer to "which model is best", and a team deploying this system should decide upfront which business objective the model is optimised for.

5. Drivers of Detraction

SHAP values were computed directly on the deployed logistic regression model, not a separate proxy, so the explanation matches what is actually in production. Global feature importance is dominated by tenure, followed by billing amount and contract type — consistent between the linear model's SHAP values and the gradient boosting model's independently-computed permutation importance, which is treated as a robustness check rather than a redundant computation.

Segment-level analysis shows that a feature's effect can fully invert depending on the customer's profile: tenure pushes toward Detractor for new customers and away from it for long-tenured ones, and the same reversal holds for contract type and internet service type. This is a reason not to act on a single global rule ("target fibre customers") without conditioning on the rest of a customer's profile.

One finding is worth highlighting as a correlation-versus-causation caveat. Customers with online security enabled show a markedly lower raw Detractor rate (14.3% versus 41.2% for those without) — but a positive SHAP contribution toward Detractor once the model accounts for tenure and contract type, which are both strongly associated with security adoption. The likely explanation is confounding, not a causal effect of the security add-on itself, and no retention action should be built on this signal alone without a controlled test.

Actionability: tenure, the strongest global driver, cannot be changed retroactively. The model's best predictor and the business's best lever are therefore not the same thing — contract type and payment method are the most defensible actionable levers this analysis surfaces, and a simple rule-based recommendation function (available in the repository) returns the top actionable driver for any given Detractor prediction, explicitly excluding non-actionable features like tenure from consideration.

6. Fairness Audit

Detractor-class recall — the share of true Detractors the model actually catches — was compared across gender, senior citizen status, and household composition (partner, dependents).

Subgroup	Detractor recall	Gap vs. overall (71.1%)
Dependents = Yes	51.4%	-19.7 pts
Partner = Yes	64.6%	-6.5 pts
Senior Citizen = No	66.4%	-4.7 pts
Gender = Female	70.4%	-0.7 pt
Gender = Male	71.9%	+0.8 pt
Dependents = No	75.8%	+4.7 pts
Senior Citizen = Yes	88.6%	+17.5 pts

Gender shows no meaningful disparity. Senior citizen status shows a real disparity, but in the opposite direction from the hypothesis raised during data exploration — senior customers are captured better, not worse. That reversal is reported as found, not adjusted to match the original expectation.

The material finding is household composition: customers with dependents are caught at 51.4% recall against 75.8% for those without — a 24-point gap comparable in magnitude to the disparity example given in the brief itself. Investigation traced this to confounding: customers with dependents have longer average tenure and far fewer month-to-month contracts, both of which the model reads as low-risk signals independent of whether the customer is actually dissatisfied. The concrete consequence is that 29% of true Detractors with dependents are misclassified as Promoters — the worst possible error — versus 15.5% for customers without dependents.

No geographic proxy audit (e.g. ZIP code as a socioeconomic proxy) was possible, since no geographic data was available to this project; this is stated as a scope gap in the audit, not implied to be covered. The recommendation is not to drop the dependents feature, which would cost real predictive performance elsewhere, but to escalate the finding to a Customer Experience or Legal review before this model is used to allocate retention budget, and to consider a differentiated decision threshold for the affected subgroup rather than a blanket model change.

7. Limitations

- The 15%-respondent business scenario cannot be fully simulated on this dataset, since Satisfaction Score is populated for 100% of customers here rather than a real 15% sample — the train/test split is stratified but not selection-bias-aware, and a production deployment would need to monitor for drift between historical respondents and the full customer base.
- No model is well-calibrated in an absolute sense; predicted probabilities should be used to rank customers, not read as literal risk percentages. The production interface displays this caveat directly to the user.
- Geographic features and their associated fairness risks (e.g. ZIP code as a socioeconomic proxy) could not be assessed — the relevant source file was not available.
- The Dependents fairness gap is documented and its likely cause investigated, but not yet mitigated with a concrete threshold adjustment or retraining strategy.
- Hyperparameter tuning was deliberately out of scope for this iteration; the evaluation effort was spent comparing model families and metrics rather than tuning a single family, consistent with a two-week time-box.

- The optional text-signal extension (synthetic customer verbatims via an LLM, embedded and combined with tabular features) has generation and evaluation scripts implemented and unit-tested, but has not yet been run end-to-end against a live LLM API to produce a reportable result.

8. Next Steps

- Run the verbatim-generation and text-signal evaluation scripts against a live LLM API to determine, with evidence rather than assumption, whether unstructured interaction notes add predictive value over the tabular baseline.
- Obtain the IBM Telco Location table to add geographic features and complete the fairness audit's coverage of geographic proxies.
- Design and test a differentiated decision threshold for the Dependents subgroup to close the recall gap identified in the fairness audit, with the trade-off against false positives made explicit to the business stakeholder approving it.
- Pin dependency versions and add a lightweight CI check that reloads the persisted model and runs a prediction smoke test — a cross-version pickling failure was encountered once during development and is a known, addressable class of production risk.
- If this model moves toward production, add a monitoring layer tracking input and prediction drift, since the respondent-versus-non-respondent distribution shift discussed above cannot be verified from this dataset alone and would need to be observed in deployment.